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Patellofemoral Pain Syndrome

Hervé Collado, MD^{a,b,c,*}, Michael Fredericson, MD^{d,e,f}

KEYWORDS

• Patellofemoral • Pain • Syndrome

The average recreational runner has a 37% to 56% incidence of being injured during the course of a year's training.¹ The knee is the most common site of injury and patellofemoral pain (PFP) syndrome constitutes nearly 25% of injuries to the knee.² The cause of PFP, however, is not clearly understood and may have multiple origins. The most commonly accepted hypothesis is related to increased patellofemoral joint stress (force per unit area) and subsequent articular cartilage wear.

CLASSIFICATION SYSTEMS

There are many classification systems for PFP. Most of these are based on radiographic findings and grading of the extent of chondral injury or patellar position. These systems are devised to help in surgical planning, but are not easily applicable in the clinical setting. An innovative classification system by Holmes and Clancy³ (**Box 1**) allows classification of the various patellofemoral pathologies seen in a runner's injury clinic, and particularly in clarifying PFP or instability related to malalignment.

PREDISPOSING FACTORS FOR PATELLAR MALALIGNMENT AND INSTABILITY

Bony Abnormalities

As the knee begins to flex, the articular surface of the patella comes into contact with the lateral femoral condyle and the patella then follows an S-shaped curve through the

^a Department of Medicine & Rehabilitation and Sport Medicine, University Hospital la Timone, Boulevard Jean Moulin, 13005 Marseille, France

^b Division of Physical Medicine & Rehabilitation, Stanford University School of Medicine, Stanford, CA, USA

^c Department of Orthopaedic Surgery, PM&R and Sports Medicine, Stanford University School of Medicine, Stanford, CA, USA

^d Orthopaedics & Sports Medicine, Stanford University School of Medicine, Stanford Center for Medicine, 450 Broadway Street, Redwood City, CA 94063, USA

^e PM&R Sports Medicine Service, Stanford University School of Medicine, Stanford Center for Medicine, 450 Broadway Street, Redwood City, CA 94063, USA

^f Stanford Cross-Country & Track, Stanford University School of Medicine, Stanford Center for Medicine, 450 Broadway Street, Redwood City, CA 94063, USA

* Corresponding author. Department of Medicine & Rehabilitation and Sport Medicine, University Hospital la Timone, Boulevard Jean Moulin, 13005 Marseille, France.

E-mail address: herve.collado@ap-hm.fr

Box 1**Classification of Patellofemoral Pain and Dysfunction***Patellofemoral instability*

1. Subluxation or dislocation, single episode
2. Subluxation or dislocation, recurrent
 - Lateral subluxation or dislocation
 - Normal functional Q-angle
 - Increased functional Q-angle
 - Dysplastic trochlea
 - Grossly inadequate medial stabilizers
 - Patella alta
 - Tight lateral retinaculum
 - Medial subluxation or dislocation
 - Iatrogenic
3. Chronic dislocation of patella
 - Congenital
 - Acquired
4. Associated fractures
 - Osteochondral
 - Avulsion

Patellofemoral pain and malalignment

1. Increased functional Q-angle
 - Femoral anteversion
 - External tibial torsion
 - Genu valgum
 - Foot hyperpronation
2. Tight lateral retinaculum (lateral patellar compression syndrome)
3. Grossly inadequate medial stabilizers
4. Electrical dissociation
5. Patella alta
6. Patella baja
7. Dysplastic femoral trochlea

Patellofemoral pain without malalignment

1. Tight medial and lateral retinacula
2. Plica
3. Osteochondritis dissecans
 - Patella
 - Femoral trochlea
4. Traumatic patellar chondromalacia
5. Fat pad syndrome

6. Patellofemoral osteoarthritis
 - Posttraumatic
 - Idiopathic
7. Patellar tendinitis
8. Quadriceps tendinitis
9. Prepatellar bursitis
10. Apophysitis
 - Osgood-Schlatter syndrome
 - Sinding-Larsen-Johanssen disease
11. Symptomatic bipartite patella
12. Other trauma
 - Quadriceps tendon rupture
 - Patellar tendon rupture
 - Patella fracture
 - Proximal tibial epiphysis (tubercle) fracture
 - Contusion
 - Turf knee/wrestler's knee
 - Cruciate ligament instability
13. Reflex sympathetic dystrophy

From Holmes WS, Clancy WG. Clinical classification of patellofemoral pain and dysfunction. J Orthop Sports Phys Ther 1998;28(5):300; with permission.

trochlea. Part of the patellar surface remains in contact with the trochlea throughout the remainder of the flexion arc. This contact between the femur and patella progresses from a distal to a proximal direction on the patella.⁴ Helping to keep the patella centered in the trochlear groove is the V-shaped anatomy of the patella and the configuration of the femoral condyles. In normal knees the lateral condyle is higher than the medial one. There may be various degrees of true dysplasia of the medial or lateral portions of the trochlear groove, leading to decreased stability of the patellofemoral joint.⁵ This instability can be caused by either excessive thickness of the floor of the trochlea or insufficient height of one or both trochlear femoral condyles. Asymmetry of patellar facets also affects patellar congruity. The normal ratio of the lateral to medial facet is 3:2, such that the lateral facet is longer and more sloped, matching the higher and wider lateral femoral condyle.⁶

Lower Extremity Malalignment

Torsional or angular malalignment of the lower extremity has a significant influence on the patellofemoral joint mechanics. Significant deviations in patella alignment secondary to patella alta, trochlear dysplasia, femoral anteversion, knee valgus, and a laterally displaced tibial tuberosity could cause PFP.⁷ For example, anteversion of the femoral neck is frequently associated with external torsion of the tibia and, frequently, compensatory pronation of the foot. Hyperpronation of the foot is a causative factor for PFP syndrome (PFPS).⁸ In theory, increased pronation causes the tibia to internally rotate during the weight acceptance phase of gait, thereby preventing the

tibia from fully externally rotating during midstance. In turn, this prevents the knee from fully locking via the screw-home mechanism. To compensate, the femur internally rotates to allow the knee to fully lock. Increased femoral internal rotation leads to increased contact pressure between the patella and the lateral trochlear groove, which can increase subchondral bone stress and symptoms of PFPS.

Muscle and Soft Tissue Imbalances

One of the most important anatomic factors proposed for affecting dynamic patellar stabilization is the balance between the medial and lateral quadriceps muscles. It is thought that the vastus medialis obliquus (VMO), the primary active medial stabilizer of the patella,⁹ is often overpowered by the lateral forces acting on the patella, which include the iliotibial tract, the lateral retinaculum, and the vastus lateralis.¹⁰ In some runners the VMO is weak; however, many runners with sufficient VMO strength still exhibit timing deficit with delayed onset of the VMO when compared with the vastus lateralis.^{11,12}

Abnormal soft tissue length can also affect patellofemoral mechanics. For example, tightness in the quadriceps muscle can directly increase the contact pressure between the articular surfaces of the femur and patella, whereas tightness in the hamstrings and gastrocnemius can indirectly increase patellofemoral joint reaction forces by producing a constant flexion moment to the patella.¹³ Tightness in the hamstrings or gastrocnemius will also restrict talocrural dorsiflexion, producing compensatory pronation in the subtalar joint¹⁴ and an increase in the dynamic Q-angle. The talocrural joint requires 10° of dorsiflexion for walking and 15° to 25° of dorsiflexion for running. If this motion is not available, compensatory pronation occurs to allow dorsiflexion of the midfoot on the rearfoot.¹⁵

Tightness of the iliotibial band (ITB) can affect normal patella excursion. The distal ITB fibers blend with the superficial and deep fibers of the lateral retinaculum, and tightness in the ITB can contribute to lateral patellar tilt and excessive pressure on the lateral patella.⁷ There is an association between patellar hypomobility and a tight ITB. Hypermobility with increased lateral patellar glide is correlated with laxity of the medial patellofemoral ligament or patellomeniscal ligament, and is often noted in association with patellar subluxation or dislocation.

PATHOPHYSIOLOGY

The pathophysiology of PFP is not well understood. Studies of the biomechanical factors that potentially contribute to PFPS demonstrate conflicting results, a lack of reproducibility, and no firm conclusions. Investigators consistently cite the difficulty of research due to the syndrome's probable multifactorial nature. Proper tracking requires balanced forces acting on the patella. If any force acting on the patella is too large or too small then the movement of the patella may be altered, thereby placing additional stresses on the soft tissue of the joint. As the stress exceeds the tissues' mechanical strength, microdamage, inflammation, and pain result.¹⁶ Altered stresses in the cartilage may also play a role in a pain response by transferring stresses into the underlying subchondral bone and exciting nociceptors. Increased cartilage and subchondral bone stress have been postulated as mechanical causes for PFP,¹⁷ supported, in part, by estimates of average joint stresses (force/area) in patients with PFP.¹⁸ The lateral retinaculum also plays an important role in PFP. The chronic lateral subluxation of the patella can lead to shortening of the retinaculum with secondary nerve damage, resembling the histopathologic picture of a Morton neuroma.^{13,19} The synovium, medial patellofemoral ligament, and fat pad of Hoffa, along with the loss of tissue homeostasis, have also been highlighted as potent sources of pain.^{20,21}

CLINICAL PRESENTATION

Runners note the insidious onset of an ill-defined ache localized to the anterior knee, behind the knee cap. The onset of anterior knee pain may be gradual or acute and may be precipitated by trauma. Pain may occur in one or both knees. Knee pain worsens with squatting, running, prolonged sitting, or when ascending or descending steps.²² The pain is often poorly localized under or around the patella, and is usually described as “achy” but may be “sharp.” The pain may vary throughout the run and is particularly aggravated by hills. Some patients may describe the affected knee as “giving way” or “buckling.” In PFPS this perceived instability may be due to pain inhibiting proper contraction of the quadriceps; but it must be distinguished from instability arising from a patellar dislocation, subluxation, or ligamentous injury of the knee. There may be occasional complaints of mild swelling but it is rare for there to be a gross effusion seen with a traumatic knee injury. In the subset of patients with patellar instability, there is often a sensation of patellar slippage or a feeling of actual bony subluxation, particularly with twisting, cutting, or pivoting activity. Only rarely, however, will a runner turn sharply enough to sustain a dynamic force sufficient to cause an acute dislocation of the patella.

PHYSICAL EXAMINATION

The positive findings in PFP are often subtle, and it is not always clear if they correlate with the patients' symptoms. Clinical studies have not been able to consistently demonstrate biomechanical or alignment differences between patients with PFP and healthy individuals.^{23–25} Some of this relates to the difficulty defining where the range of normal alignment ends and malalignment begins. Physical examination for static malalignment has a place in clinical and scientific evaluation. However, many of these malalignments change once movement is initiated. Furthermore, additional malalignment may exist during movement as a result of poor muscular control of the segments.¹⁶ Given this, the authors still believe that the physical examination, when systematically performed, can highlight predisposing static and dynamic factors to patellofemoral malalignment that are important to address in the design of an individualized treatment program.²⁶

Standing Examination

Static alignment

The anatomic alignment of the pelvis and lower extremity may play a role in the development of PFPS in some individuals, and is referred to as “static alignment” because it is identifiable when the patient is not moving. Static alignment is not easily modified with conservative rehabilitation. Alignment of the lower extremity is evaluated by noting evidence of femoral anteversion; knee position (genu varum, valgum, or recurvatum); external tibial rotation; and foot and ankle weight-bearing alignment. A common clinical measurement of the alignment of the lower extremity is the Q-angle. The Q-angle is formed by the line connecting the anterosuperior iliac spine to the center of the patella and the line connecting the center of the patella to the middle of the anterior tibial tuberosity. This angle is thought to represent the line of action of the quadriceps force. Although many sources report that a Q-angle greater than 16° is a risk factor for developing PFPS, the literature is inconsistent in supporting this finding.^{16,24} Although some investigators^{27,28} have found a difference between the Q-angles of men and women, a study by Grelsamer and colleagues²⁹ concluded that when measurement data are corrected for the average height between men and women, the difference in the Q-angle between men and women disappears.

Dynamic alignment

Dynamic malalignment may exist during movement as a result of poor muscular control of the lower extremity segments. Dynamic alignment can be tested by having the runner step slowly upon and down from a 6-inch stool or perform single-leg squats. The presence of any abnormal movements of the patella as it engages and disengages the trochlea, and any body shifting, trunk rotation, or loss of hip control are noted. The “dynamic Q-angle” that causes dysfunction has been described. The concept of poor dynamic alignment took form when clinicians observed a consistent pattern when patients performed a single-leg squat or step-down of excessive contralateral pelvic drop; hip adduction and internal rotation; knee abduction; and tibial external rotation and hyperpronation. Several investigators have described this movement pattern and its link to PFPS.^{30–32} Dynamic magnetic resonance imaging (MRI) has shown that during weight bearing, PFPS subjects had greater femoral internal rotation than control subjects.³³

This concept supports the idea that poor control of the femur may contribute to PFPS symptoms in some patients. The excessive frontal and transverse plane motion seen in women during functional tasks has been attributed to weakness in the hip abductors and external rotators.^{30,31,34,35} Gluteus medius weakness allows the contralateral pelvis to drop, and puts the stance leg in an adducted position; this is accompanied by excessive internal rotation of the femur and tibia and hyperpronation of the subtalar joint.^{36,37} Normally the patella travels smoothly through the trochlear groove and follows a straight or slightly curved path. An abrupt or sudden lateral movement of the patella as the knee nears full extension is considered abnormal. Called a positive J-sign, this movement is caused by excessive lateral forces as the patella exits the femoral trochlea at 10° to 30° of flexion,^{6,38} and is seen in a small number of patients with patellar malalignment and in the majority of patients with frank patellofemoral instability.

Sitting Examination

In the sitting position, patellar height for signs of alta or baja position can also be assessed. A prominent infrapatellar fat pad often accompanies patella alta, as does genu recurvatum. Patella alta is more frequently seen in women and is a common finding in a congenitally subluxing patella, as it causes the patella to enter the femoral sulcus late in knee flexion. Patella baja is rarer and is sometimes seen as a complication of anterior cruciate ligament reconstruction. The path of the patellar tendon insertion into the tibial tuberosity is also noted. When the tibial tubercle is situated more laterally than normal, such that the patellar tendon descends at an angle rather than directly downward, proximal external tibial torsion is present.

Dynamic patellar tracking is a measure of patellar instability. During this evaluation, the examiner asks the seated patient to actively extend the knee from 90° to full extension, and observes the movement pattern of the patella from the front. In most individuals, the patella seems to move straight proximally, with a slight lateral shift near terminal extension (J-sign). The term J-sign describes the path of the patella with mal-tracking. Instead of moving superiorly with knee extension, the patella suddenly deviates laterally at terminal extension as it exits the trochlear groove, to create an inverted J-shaped path.^{22,39}

Supine Examination

Leg length discrepancies can be screened by measuring the distance from the anterior superior iliac spine to the highest point of the medial malleolus. Leg length discrepancy greater than 1.0 cm may have adverse effects on the lower extremity while

running.⁴⁰ The knee joint is then observed and palpated for any swelling. As little as 20 to 30 mL of fluid in the knee joint can inhibit VMO function.⁴¹ It is unusual, however, to have any more than a mild synovitis of the knee with chronic extensor mechanism problems.

The examination then focuses on palpation of the patella and peripatellar soft tissues for tenderness. The lateral retinaculum interdigitates with fibers from the vastus lateralis and ITB, and tenderness is frequently palpated in this area related to the chronic, recurrent stress of a maligned patella.^{38,42} Several studies have presented evidence of nerve damage and hyperinnervation into the lateral retinaculum in patients with patellofemoral malalignment.^{43,44} In these individuals, the neural growth factor is overexpressed in the nerve fiber and vessel wall, and stimulates the release of substance P in the free nerve endings.⁴⁴ This condition is often associated with tenderness along the medial patellar facet, and less often, the lateral patellar facet. These areas are best palpated by curling the fingers around the border of the patella. The examiner next palpates along the patellar tendon, specifically at its attachment to the distal pole of the patella. Patellar tendinitis is demonstrated by tenderness to palpation in this area. Tenderness directly ventral or dorsal to the patellar tendon is associated with patellar bursitis and tendinopathy, whereas tenderness along either side of the patellar tendon is associated with inflammation of the fat pad of Hoffa. Less commonly, tenderness is located at the proximal pole of the patella at the quadriceps insertion, indicating quadriceps tendinitis.

Close observation of the patella in relation to the femur is continued. Grelsamer and McConnell¹⁰ describe 4 components that are believed to affect patellar position statically or dynamically: glide, tilt, rotation, and anterior-posterior position. The glide test is an assessment of lateral/medial displacement of the patella, and measures the distance from the midpole of the patella to the medial and lateral femoral epicondyles with the knee flexed to 20°. The patella may sit equidistant to the condyles, but moves laterally when the quadriceps contract, indicating a dynamic problem. The glide component is examined by use of a tape measure to record the distance from the midpatella to the lateral femoral epicondyle and the distance from the midpatella to the medial femoral epicondyle. The midpatella point is determined by visual assessment.¹⁰ A 5-mm lateral displacement of the patella caused a 50% decrease in vastus medialis oblique tension.⁴⁵ While it is common for patients with PFP to have some degree of lateral displacement, these clinical measurements should be interpreted with caution, because they may not be a true reflection of anatomic position. Powers and colleagues⁴⁶ found that the clinical assessment of patellar glide (medial/lateral displacement) overestimated the true amount of lateral displacement assessed on MRI.

Patellar tilt compares the height of the anterior aspect of the medial patellar border with the height of the anterior aspect of the lateral patellar border. This tilt is considered normal when the 2 borders are level in the frontal plane. In mild tilt, more than 50% of the depth of the lateral border can be palpated but the posterior surface is not palpable; and in more severe tilt less than 50% of the depth of the lateral border can be palpated. To determine the presence of a dynamic tilt problem, an active contraction is simulated by passively moving the patella medially. If the depth of the lateral border becomes more difficult to palpate, a dynamic tilt is present. If the lateral tilt is severe, this can lead to lateral patellar compression syndrome, sometimes requiring surgical release (**Fig. 1**).

Rotational measurements help to determine any deviation of the long axis of the patella from the long axis of the femur, and are believed to be another indication that a particular part of the retinaculum is tight and a potential source of symptoms.

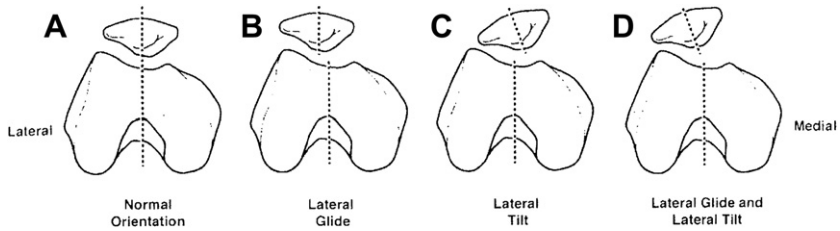


Fig. 1. Example of patellar orientation with a lateral glide and/or tilt component.

In normal rotation a line connecting the superior and inferior poles of the patella is parallel to the long axis of the femur. If the inferior pole is medial to the long axis of the femur, this signifies internal rotation; if the inferior pole of the patella is lateral to the long axis of the femur, this signifies external rotation.

The evaluation of anteroposterior (A-P) alignment is to assess if the inferior pole is tilted posteriorly compared with the superior pole. Such tilting can irritate the fat pad and is common in patients who have pain on extension or hyperextension of the knee, because the inferior pole gets buried in the fat pad. These patients are often diagnosed as having a patellar tendonitis, and usually have pain with quad sets and straight leg raises. A-P tilt occurs when the distal one-third of the patella, or the inferior pole of the patella, is not as clear to palpate as the superior one-third and superior pole. These individuals often present with a dimple in their knees. Dynamically anterior-posterior tilt can be determined during a maximal quadriceps contraction. If the inferior pole disappears as a dimple, then the test is positive.

Also important is assessment of patellar mobility.⁴⁷ The test is performed with the knee flexed 20° to 30° and the quadriceps relaxed. The test can be done either by resting the patient's knee over the examiner's thigh or with a small pillow underneath the patient's knee. The patella is divided into 4 longitudinal quadrants, and the patella is displaced medially and laterally with the examiner's thumb and index finger to determine the amount of patellar tightness (**Fig. 2**). A lateral displacement of 3 quadrants suggests an incompetent medial restraint. A lateral displacement of 4 quadrants defines a dislocatable patella. A medial displacement of only 1 quadrant indicates a tight lateral retinaculum and usually correlates with an abnormal passive patellar tilt test. There is an association between patellar hypomobility and a tight ITB.⁴⁸ Medial displacement of 3 or 4 quadrants suggests a more globally hypermobile patella without tightness of the lateral restraints, and is often seen in patients with other stigmata of generalized ligamentous laxity. Hypermobility with lateral patellar glide is correlated with laxity of the medial patellofemoral ligament or patellomeniscal ligament, and is often noted in association with patellar subluxation. During these maneuvers, the examiner should look not only for patellar mobility but also for any associated apprehension. When positive, this is a very specific test for patellar instability.

The ligamentous stability of both knees is assessed, particularly if there is a previous history of knee injury. Both anterior and posterior cruciate deficiencies are associated with peripatellar pain. Careful evaluation for meniscal pathology is noted by palpation of the medial and lateral joint lines and McMurray testing. Joint-line tenderness can be present with patellar pathology if there is an associated synovitis, and does not necessarily indicate a meniscal injury or femorotibial arthritis.¹⁰ Identification of a symptomatic synovial plica is equally important.

Hip range of motion, femoral acetabular impingement tests, and the Faber maneuver should be thoroughly assessed to rule out referred pain to the knee from

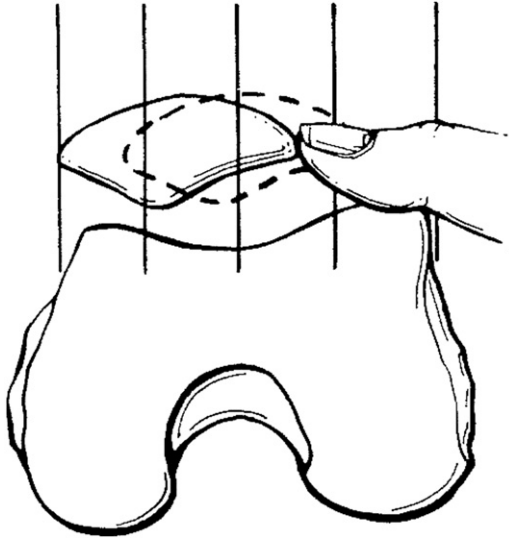


Fig. 2. Assessment of patellar mobility medially and laterally. (*From Assessment of patellar mobility medially and laterally. In: DeLee JC, Drez D, editors. Orthopaedic sports medicine, vol. 2. Philadelphia (PA): W.B. Saunders Co; 1994. p. 1179; with permission.*)

intra-articular hip pathology. Internal rotation that exceeds external rotation is suggestive of femoral anteversion, and following this, soft tissue length should be measured in the hamstrings and hip flexors.

Side-Lying Examination

In the side-lying position, with the knee flexed at 20° , the lateral retinaculum can be evaluated for excessive tightness by passively moving the patella in a medial direction.¹⁰ The superficial retinacular fibers are believed to be tight if the femoral condyle is not easily exposed. To test the deep fibers, the hand is placed on the middle of the patella, the slack of any lateral glide is removed, and an anteroposterior pressure on the medial border of the patella is applied. The lateral border of the patella should move freely away from the femur, and on palpation the tension in the retinacular fibers should be similar.

In this position ITB tightness can be evaluated with Ober's test, and the gluteus medius can be tested for strength deficits.

Prone Examination

This position allows a more accurate assessment of rearfoot and forefoot alignment, subtalar position, gastroc-soleus, and quadriceps muscle length.

Observational Gait Analysis

Observational gait analysis is one of the most important aspects of the examination—the evaluation of dynamic function by observing the runner's angle of gait while walking and running. In more complicated cases, it may be helpful to use treadmill running or even more sophisticated video taping to analyze the runner's gait. The findings of Thijs and colleagues⁴⁹ suggest that an excessive impact-shock during heel-strike and at the propulsion phase of running may contribute to an increased risk of developing PFP. Dierks and colleagues⁵⁰ have also shown that runners with PFPS

displayed weaker hip abductor muscles that were associated with an increase in hip adduction during running. This relationship became more pronounced at the end of the run.

DIAGNOSTIC STUDIES

The diagnosis of PFPS depends primarily on the history of the patient. Radiography is an adjunct to history and physical examination. It is important to obtain radiographs in the runner who has apparent PFP and does not demonstrate improvement after several weeks of conservative treatment, or if there has been a history of recent trauma or dislocation. Radiographic findings do not correlate well with clinical complaints and frequently the injured side is difficult to differentiate from the uninjured side.⁵¹ When imaging is indicated, plain films of the knee (weight-bearing A-P, weight-bearing lateral, and axial view) are useful to rule out other sources of anterior knee pain, including bipartite patella, osteoarthritis, loose bodies, and occult fractures.

On standard A-P radiographs of the knee, one can identify accessory ossification centers, degenerative joint disease, and other unrelated conditions, such as bone tumors.⁵²

The lateral view is most helpful for assessing patellar height. There is a plethora of measurement techniques described in the literature for this purpose.¹⁰ The Blackburn and Peel⁵³ technique is easy to use and fairly reliable. This technique measures the distance from the tibial plateau to the inferior pole, which should equal the length of the patellar articular surface (**Fig. 3**). Normal values are approximately 1.0, with higher values indicating patellar alta. The technique should be used with caution in adolescent patients with skeletally immature proximal tibial epiphysis, certain patellar morphotypes with short articular surface, or when there is an abnormal slope of the tibial plateau.¹⁰

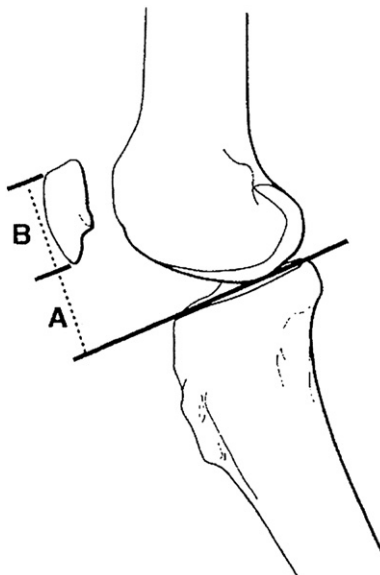


Fig. 3. Technique of Blackburn and Peel to assess patella alta on lateral radiographs of the knee.

Axial or Merchant views of the patellofemoral joint are recommended with the knee flexed 30° .¹⁰ Some subluxation can be detected at 30° that could be missed at 45° . This view allows evaluation of degenerative changes in the patellofemoral joint, osteochondritis dissecans of the patella, patellar morphology, dysplasia of the trochlear groove, and accessory ossification centers and ectopic calcifications in the retinaculum. The position and orientation of the patella relative to the trochlear groove are also evaluated with the sulcus angle, congruence angle, and the patellar tilt angle.

The sulcus angle measures the angle of the bony trochlea. With the knee flexed at 30° to 45° , the normal sulcus angle is approximately 140° .⁵⁴ Patellar instability is associated with a more shallow trochlea, whereas too steep a trochlea is associated with patellar pain without instability.⁵⁵

The congruence angle is an index of medial/lateral subluxation of the patella within the trochlear groove, similar to the physical examination assessment of patellar glide, and in nondislocators the average angle is -6° (standard deviation 11°) (Fig. 4A).⁴² Although this measurement has been widely used, drawing and bisecting an angle can prove to be time-consuming and difficult with both plain radiographs and digital radiography. A new axial linear displacement measurement presented in the study by Urch and colleagues⁵⁶ provides a simple alternative method for evaluating the position of the patella relative to the trochlear groove that is similar to that of the congruence angle measurement (see Fig. 4B).

The patellar tilt angle is an index of the medial/lateral tilt of the plane of the patella relative to the femur (Fig. 5). In the normal patellofemoral joint, the angle formed by the lateral patellar facet and any horizontal line should open laterally, whereas in patients with patellar subluxation the lines used to define the angle are parallel or open medially.⁵⁷ For this evaluation to be accurate and repeatable, the radiograph must be taken with the foot vertical and the cassette maintained parallel to the ground. If the patient exhibits external tibial torsion, the natural position of the feet is maintained. A tilt angle between 0° and 5° is normal, 5° to 10° is borderline, and an angle greater than 10° is considered abnormal. In a study by Grelsamer and colleagues⁵⁸ abnormal tilt was detected in 85% of patients suffering from malalignment pain. The 15% of patients whose malalignment was not detected exhibited either abnormal tilt that became normal at 30° of flexion or malalignment that was not related to tilt (ie, patella alta or lateral displacement).

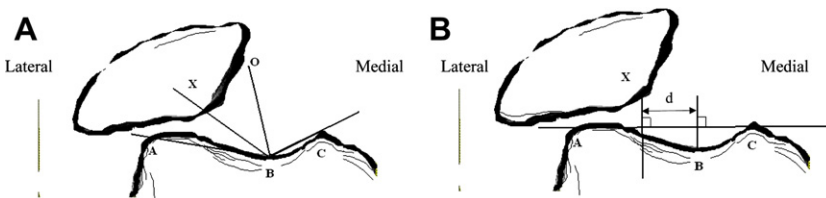


Fig. 4. (A) Congruence angle. Line BO is the bisector of angle ABC. Line BX passes through the lowest point on the median ridge of the patella. Angle OBX is the congruence angle. If line BX falls to the medial side of line BO, the angle is expressed as negative degrees. If it falls to the lateral side of line BO, it is expressed as positive degrees. (B) A reference line is drawn by connecting the most anterior aspects of the lateral (A) and medial (C) trochlear facets. A line is drawn perpendicularly from the depth of the sulcus (B) through the reference line (AC). Another line is drawn perpendicularly from the posterior aspect of the patellar spine (X) through the reference line. To obtain the lateral displacement measurement, the distance (d) between the 2 intersections is measured.

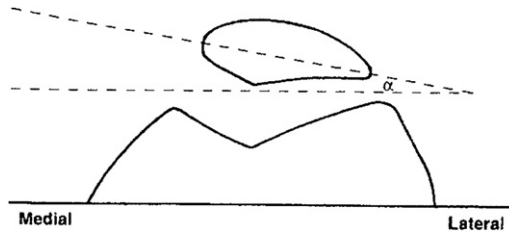


Fig. 5. Patellar tilt angle. The tilt angle is measured by a line joining the corners of the patella and any horizontal line. (From Grelsamer RP, McConnell J. The patella. Gaithersburg (MD): Aspen Publishers; 1998. Adapted from Grelsamer, RP, Bazos AN, Proctor CS. Radiographic analysis of patellar tilt. J Bone Joint Surg Br. 1993 Sep ;75(5):822–4; with permission.)

Because the patella usually becomes unstable as it nears extension, a lateralized patella may not be detected by an axial-view radiograph that requires the knee be flexed to at least 30°. Thus if surgery is contemplated and plain films are negative, an MRI scan or computed tomography (CT) scan should be used to further evaluate patellar tracking. Serial CT scans taken at knee flexion angles in the range of 0° to 30° can provide even greater information about the tracking of the patella in the trochlear groove. Measurements of sulcus angle, congruence angle, and patellar tilt angle can be performed on CT scans just as they can be performed on conventional axial radiographs, and may be helpful in understanding the patient with a difficult patellofemoral problem.^{59,60} Three patterns of malalignment can be characterized by the congruence angle and patellar tilt angle: lateral subluxation without lateral tilt, subluxation with tilt, and tilt without subluxation.

Kinematic studies can also be accomplished with MRI^{61,62} without exposure to radiation. In addition to defining tracking abnormalities, MRI is particularly helpful in assessing any degenerative joint changes, such as cartilage fissuring or thinning, subchondral bone marrow edema, subchondral cysts, and other pathologic entities such as synovial plica and patellar tendinitis. A new method using real-time MRI allows measurement of patellofemoral joint motion during weight-bearing and non-weight-bearing dynamic tasks (Fig. 6).⁶³ The aim is to define the cause of PFP using real-time MRI in experimental research. Single-slice, spiral, real-time MR images of all subjects performing knee flexion/extension were obtained in a 0.5-T open MRI scanner (Fig. 7).

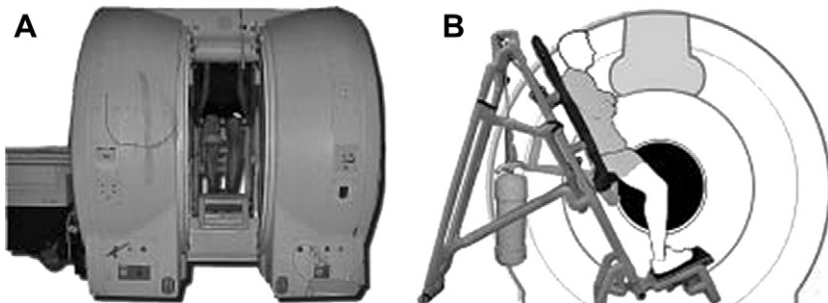


Fig. 6. (A) Open-bore MRI scanner with subject performing a weight-bearing squat. (B) Side view of scanner and backrest used to stabilize subjects as they moved in the scanner. Subjects supported about 90% of their body weight.

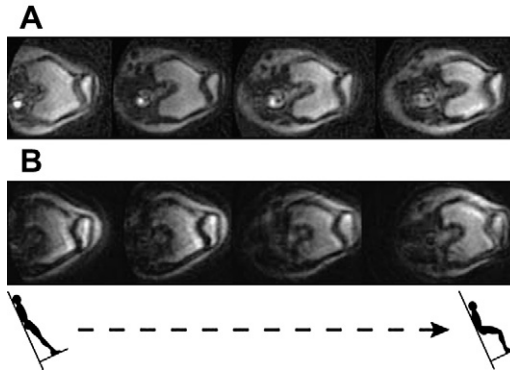


Fig. 7. (A) Real-time MR images of the patellofemoral (PF) joint of a pain-free control subject during upright, weight-bearing knee extension. (B) Real-time MR images of the PF joint of a subject with pain during upright, weight-bearing knee extension. Notice the lateral position and rotation of the patella relative to the femur.

REHABILITATION

The management of PFPS should include a comprehensive rehabilitation program. Symptom control (activity modification, nonsteroidal anti-inflammatory drugs, ice or cold application, patellar taping, or patellar bracing) is the first stage, then patients should be classified by suspected contributing mechanisms: abnormal patellofemoral joint mechanics, altered lower extremity alignment and/or motion, and overuse.⁷

Abnormal Patellofemoral Tracking and Alignment

Tightness of soft tissue structures

Soft tissue inflexibilities, particularly in the hamstrings and ITB, can also affect normal patella excursion and should be addressed.^{7,64} Soft tissue mobilization and stretching techniques are also important to address tight retinaculum structures contributing to abnormal patellar tracking.

Decreased patellar mobility

Patellar mobilization techniques should be performed if evidence of decreased patellar mobility is noted. These techniques should be performed with care to prevent excessive patellofemoral joint compression.

Quadriceps muscle strengthening

Restoration of quadriceps strength and function has been demonstrated to be a significant contributing factor to recovery from patellofemoral symptoms.⁶⁵ However, the mechanism by which strengthening improves PFP symptoms and functional ability is not entirely clear.

Choosing the correct exercises to prescribe for an individual with PFP requires an understanding of patellofemoral joint biomechanics. During open chain exercises, the amount of quadriceps muscle force required to extend the knee steadily increases as the knee moves from 90° to full knee extension.¹¹ In addition, the patellofemoral joint contact area decreases as the knee extends, thereby increasing patellofemoral joint stress. By contrast, during closed chain exercises the quadriceps muscle force is minimal at full knee extension and therefore, patellofemoral joint stresses are

reduced.¹¹ Examples of closed chain exercises include lunges, wall slides, and leg press machines.

Both open and closed chain strengthening exercises should be performed so that strengthening can be performed throughout a large arc of motion. Isometrics and open kinetic chain exercises, such as knee extensions, are recommended if there is significant quadriceps weakness or pain with weight bearing.⁶⁶ As quickly as possible, however, patients should be progressed to the core of the rehabilitation program that includes a closed kinetic chain quadriceps strengthening regimen, which is more effective than isokinetic joint isolation and open chain exercises in improving function.^{67,68}

To improve eccentric control of the quadriceps, the rehabilitation program also should include exercises performed while standing on one leg. In this position, the lower abdominals and the gluteals work together to maintain a level pelvis, simulating the activity of the stance phase of gait.¹⁰ Activation of the lower abdominal and oblique muscles helps to decrease the anterior rotation of the pelvis and the resultant internal rotation of the femur (step-down exercises: **Fig. 8**).

In addition, Souza and Powers⁶⁹ have demonstrated abnormal hip kinematics in women with PFP that seem to be the result of diminished hip-muscle performance rather than altered femoral structure. The results suggest that assessment of



Fig. 8. Step-down exercise. The pelvis should remain parallel with the floor; the hip, knee and foot should be aligned, avoiding excessive hip adduction or internal rotation. The goal is to activate the gluteal and quadricep muscles by maintaining erect posture, and avoiding forward body lean as one steps down. Start on a step only a few inches off the ground and increase the distance as stability improves.

hip-muscle performance should be considered in the evaluation and treatment of patellofemoral joint dysfunction.

Role of the Vastus Medialis Obliquus

Isolated recruitment of the VMO, proposed by many therapists, has not been proven to occur with exercises that are commonly prescribed for the treatment of PFP. One electromyographic study examined 9 commonly used strengthening exercises and found that the activity of the VMO was not significantly greater than that of the vastus lateralis, the vastus intermedius, and the vastus medialis longus, suggesting that isolated recruitment or strengthening of the VMO through selected exercises is unrealistic and probably translates into a general quadriceps muscle strengthening effect.^{7,70}

Patellar Taping

Correcting abnormal patella posture using the Grelsamer and McConnell¹⁰ taping technique is one way of optimizing the entry of the patella into the trochlea, and is a transitional step in the rehabilitation process in those patients who are unable to perform strengthening exercises due to their pain. Taping the patella of symptomatic individuals during stair ascent and descent exercises can diminish symptoms by 50%. In addition, it has been shown to increase quadriceps activity and permit increased loading of the knee joint.¹¹

Patellar Braces

Patients with patellar pain may report decreased pain from wearing properly fitted dynamic patellar stabilization braces. Powers and colleagues⁷¹ found that 50% of subjects experienced an improvement in symptoms with the use of patellar stabilizing braces. Further study⁶⁴ evaluating the same brace, however, did not find it was able to consistently correct patellar tracking patterns as measured quantitatively by kinematic MRI. However, a more recent study by Draper and colleagues⁶³ evaluated women with PFP during dynamic, weight-bearing knee extension, and assessed the effects of knee braces on patellofemoral motion with real-time magnetic resonance images of the patellofemoral joints of 36 women volunteers. The results suggest that different subsets of women with PFP exist and that bracing seems to have a mechanical effect in only one group compared with the controls: those with abnormal kinematics. With application of the brace, the lateral translation and tilt of the patella were reduced but were not restored to normal. A simple patellar sleeve provided more mild correction of lateral translation of the patella but did not alter patellar tilt.

Lower Kinetic Chain Problems

Subtalar joint pronation

Excessive pronation whether secondary to abnormal compensation as a result of abnormal structure in the trunk or lower extremity, or secondary to pathology in the foot itself, should also be addressed. Excessive pronation can increase the Q-angle, causing a dynamic abduction moment at the knee and a consequent increase in the laterally directed force on the patella. It should be noted that not all patients with excessive pronation need an orthotic. In fact, a recent study⁷² showed that although foot orthoses are superior to flat inserts according to participants' overall perception, they are similar to physiotherapy and do not improve outcomes when added to physiotherapy in the short-term management of PFP.

In those runners who seem to benefit from an orthotic, a custom-molded orthotic for maximal biomechanical control is recommended. Most over-the-counter orthotic devices have been designed to address primary control of rearfoot motion.

Three-dimensional studies, however, have shown that forefoot stability may play an integral role in rearfoot stability.⁷³ These studies indicate that instability in the forefoot at push-off may create instability in the rearfoot. For this reason an orthotic that extends all the way to the sulcus or web-space of the toes is considered crucial for control of forefoot instability in runners.¹⁰

Hip internal rotation

The functional significance of an internally rotated femur is that the trochlear groove can rotate beneath the patella, placing the patella in a fairly lateral position. If it is observed that the femur collapses into internal rotation during gait, and this motion seems to originate from the pelvis, then strengthening of the external rotators including gluteus maximus, gluteus medius, and the deep rotators may be indicated. If the femur remains in a constant state of internal rotation during the gait cycle (rather than collapsing inward), femoral anteversion should be suspected. As femoral anteversion is a fixed bony deformity, little can be done from a nonsurgical standpoint to correct this abnormality.

Gait Deviations

The restoration of normal gait function is essential to the overall treatment plan. Preliminary work from Noehren and Davis⁷⁴ suggests that gait retraining with real-time visual feedback can alter faulty mechanics, such as increased hip adduction and internal rotation and pelvic drop. Research in cadavers has shown that decreasing hip adduction or internal rotation decreases the compression on the lateral facet of the patella.^{75,76} In addition, decreased pelvic drop reduces the strain on the ITB, which in turn reduces its lateral force on the patella and allows more normal alignment.⁷⁷

Overuse

The lack of significant findings on physical examination (ie, normal patellar mechanics, normal lower extremity function) suggests that the source of PFP symptoms may be related to overuse. This overuse is often seen in an athletic population. The treatment program should focus on relative rest with activity modification. The training program also should be evaluated for obvious errors, including increasing exercise intensity too quickly, inadequate time for recovery, and excessive hill work.

Most runners with PFP and malalignment, including those with minor instability problems, will respond to nonoperative treatment. The lateral retinacular release is perhaps the most frequently performed surgical procedure, but is only indicated if a major contributor to malalignment is a tight lateral retinaculum causing isolated patellar tilt.⁴⁷ A common complication of lateral retinacular release is an incomplete release, which most commonly is the failure to release the patellar tibial portion of the lateral retinaculum along with the patellofemoral portion. By contrast, an overzealous release can lead to medial subluxation or dislocation. A lateral release will not correct more global patellar hypermobility or an abnormal anatomic Q-angle. In these cases, more extensive realignment surgery may be contemplated but is unlikely to permit a return to serious running.^{14,42}

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